Cell-type specific network of long-range Polycomb loops steers neuronal subtype identity

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1. Medical University of Vienna, Austria
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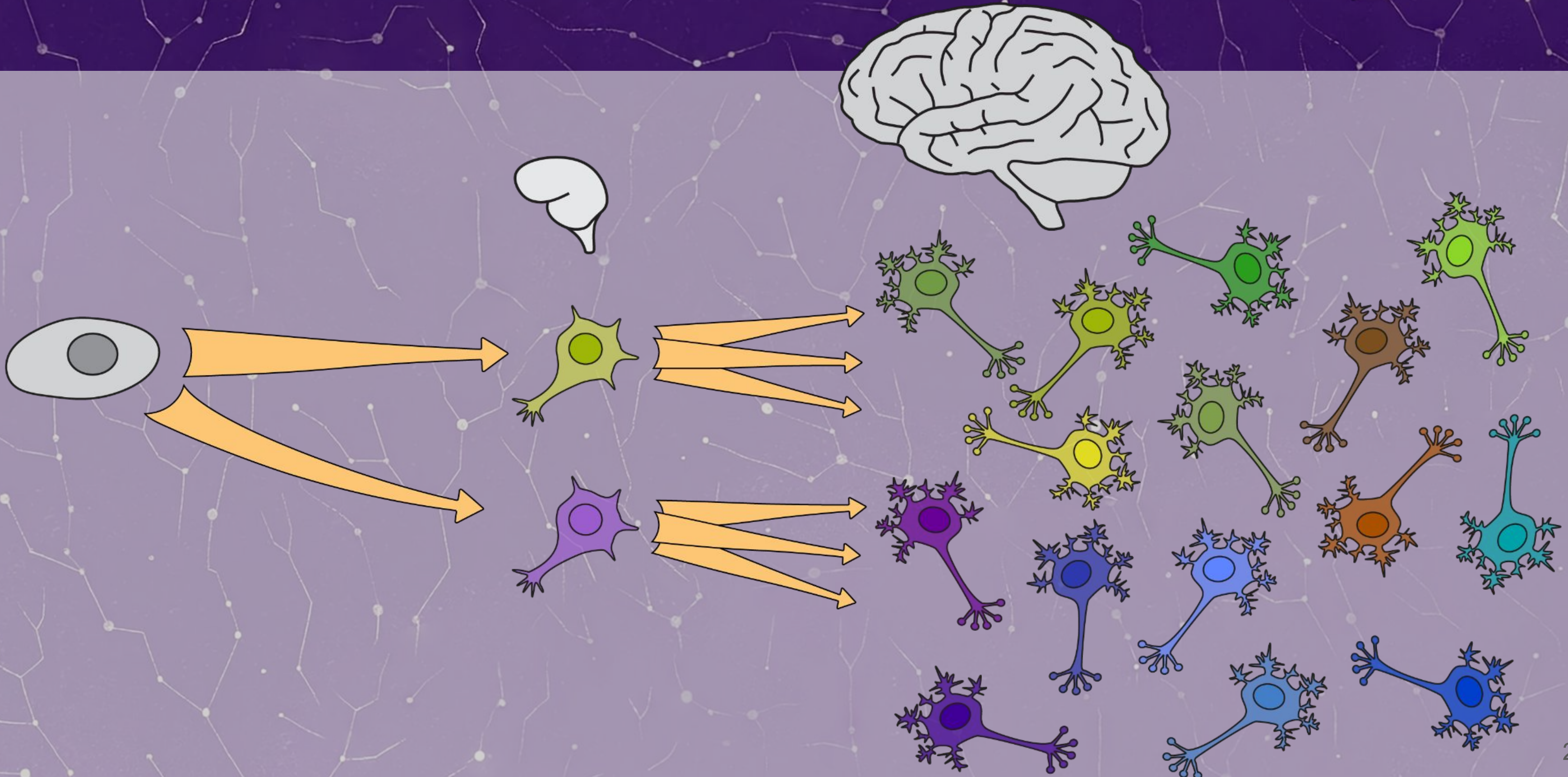
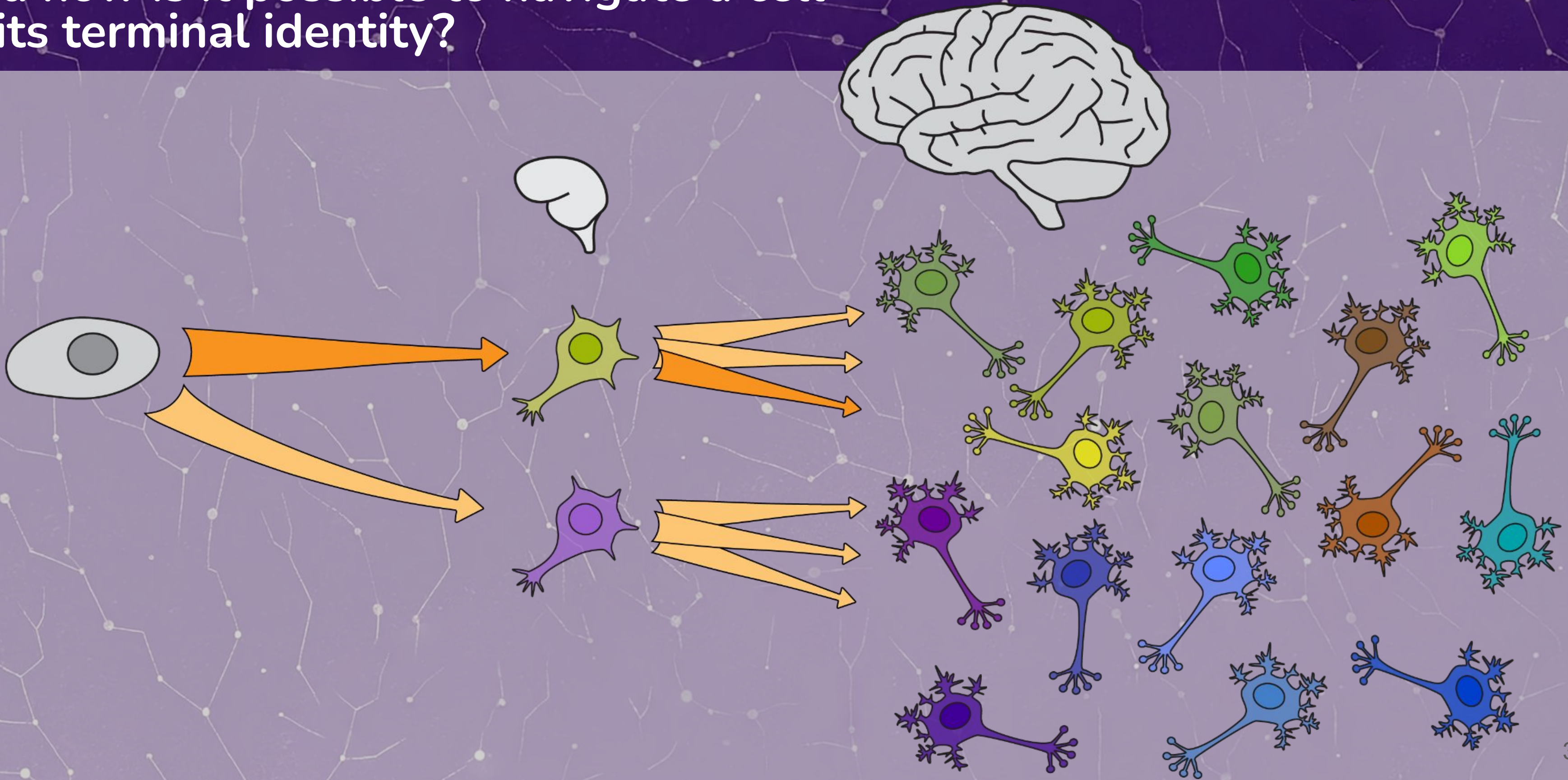
The cosmos of neuronal subtypes is tremendously large



The cosmos of neuronal subtypes is tremendously large

And how is it possible to navigate a cell to its terminal identity?



The cosmos of neuronal subtypes is tremendously large

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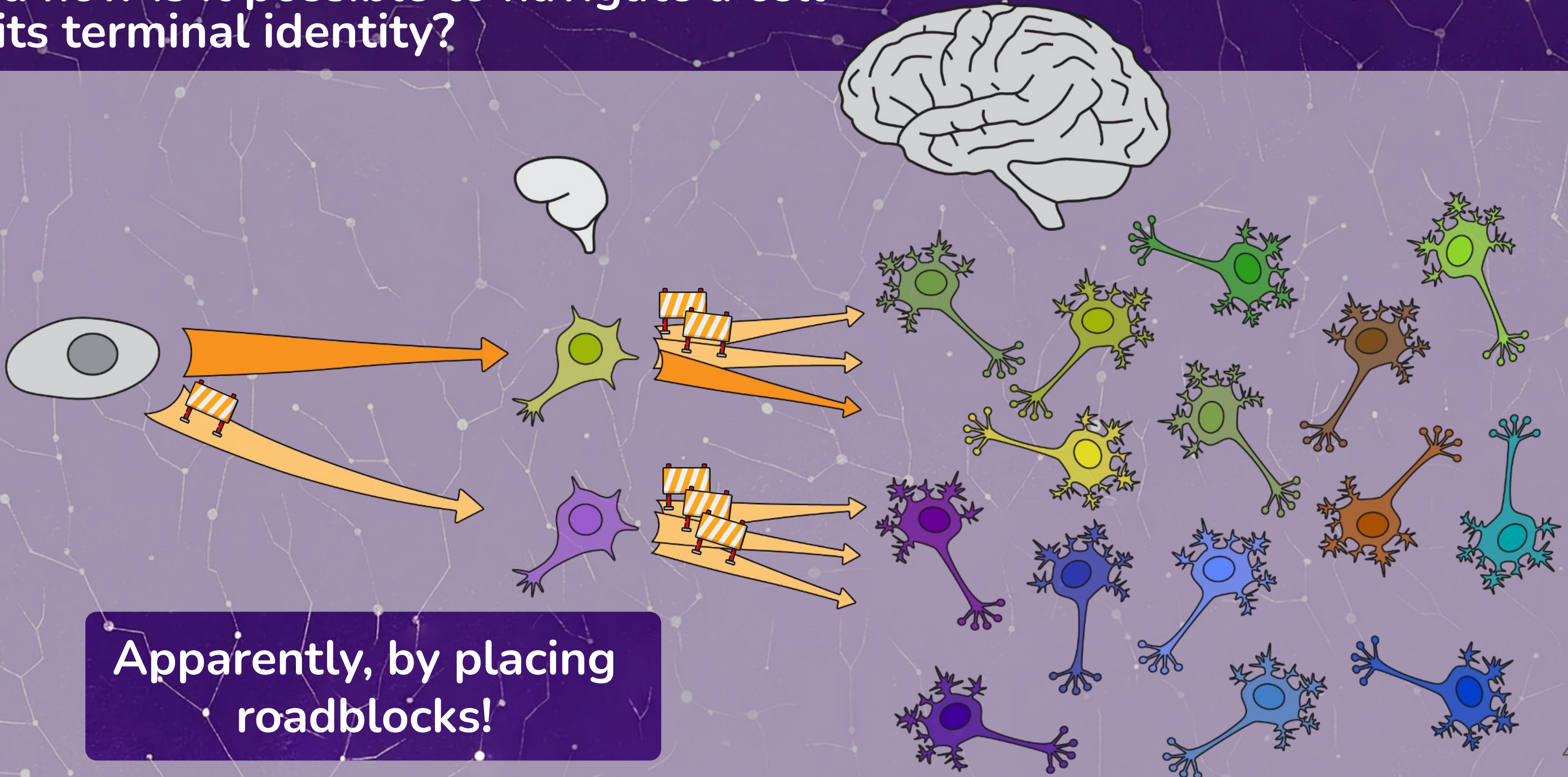
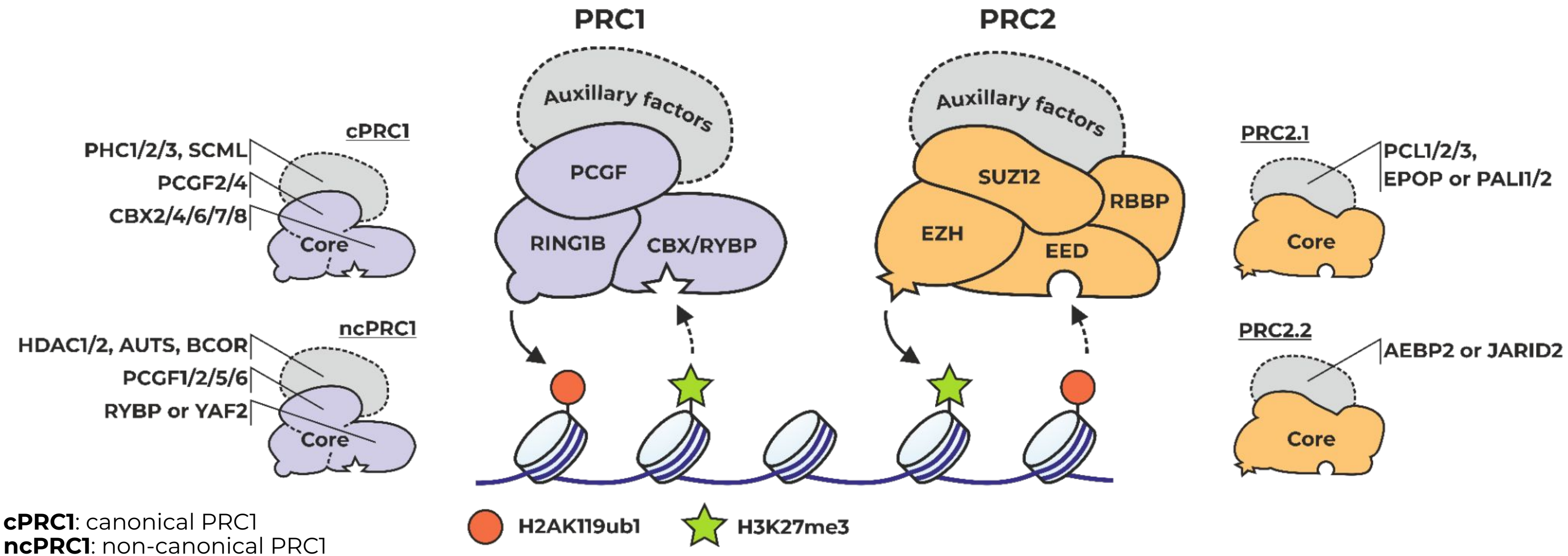


Apparently, by placing roadblocks!

Polycomb as a roadblocking machinery

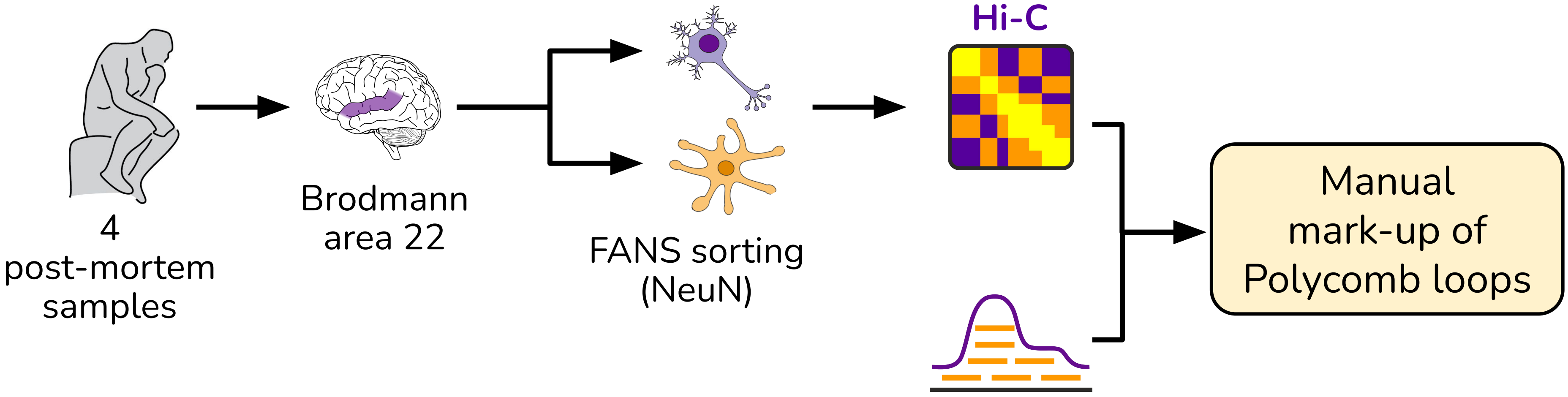
Polycomb group (PcG) proteins form two multisubunit **repressive** complexes (**PRCs**) with functions as:

histone modification, ***gene silencing***, ***chromatin structuring***.



cPRC1: canonical PRC1
ncPRC1: non-canonical PRC1

Mapping Polycomb loops



From our previous paper: NAR 2024

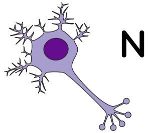
JOURNAL ARTICLE

Extensive long-range polycomb interactions and weak compartmentalization are hallmarks of human neuronal 3D genome

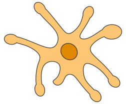
Ilya A Pletenev, Maria Bazarevich, Diana R Zagirova, Anna D Kononkova, Alexander V Cherkasov, Olga I Efimova, Eugenia A Tiukacheva, Kirill V Morozov, Kirill A Ulianov, Dmitriy Komkov ... Show more

*P. Dong, ..., P. Roussos, Nat. Genet, 2022

Polycomb loops in human brain 3D genome

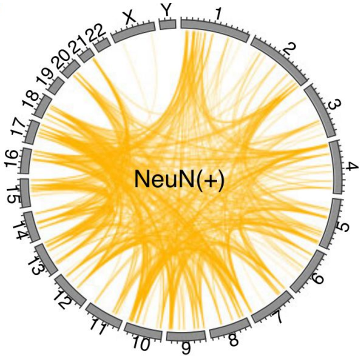


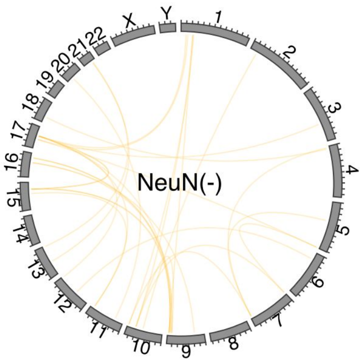
Neurons



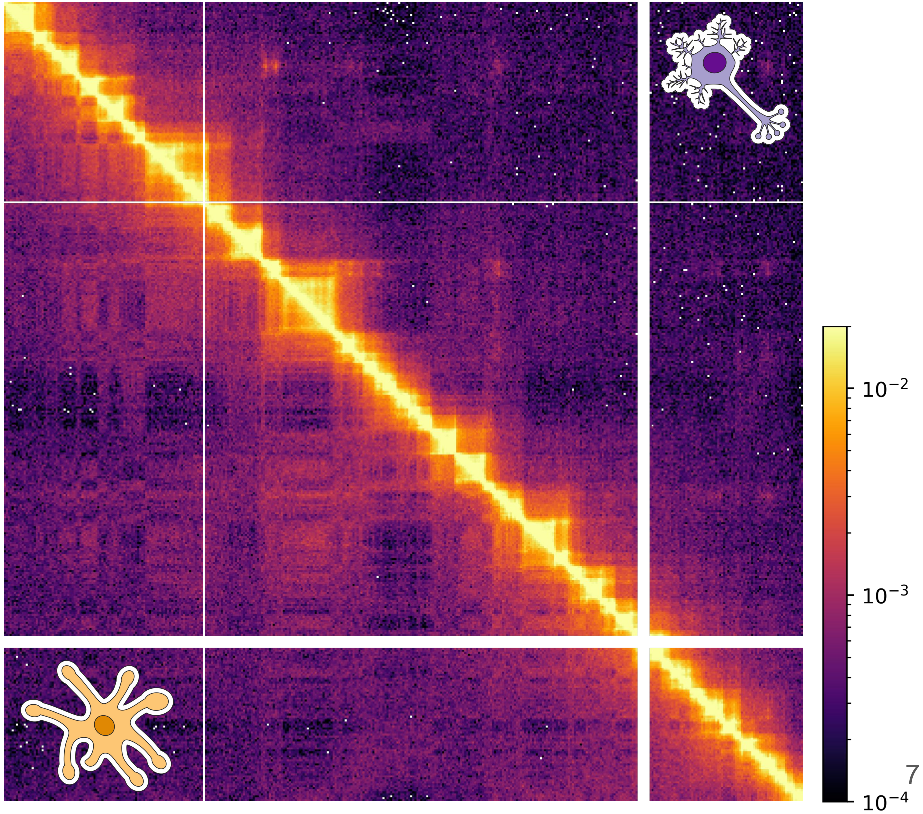
Glia

Polycomb makes bright loops in **neuronal** 3D genome

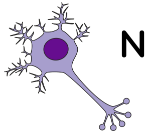




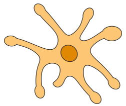
chr 6, 80 Mb - 140 Mb



Polycomb loops in human brain 3D genome

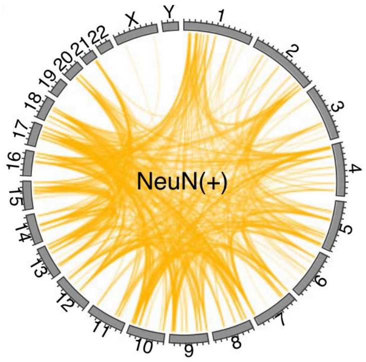


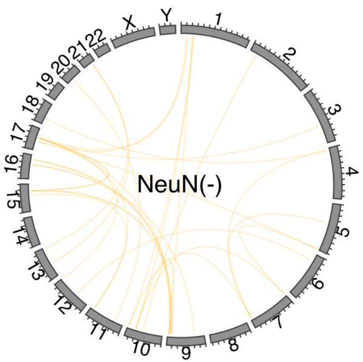
Neurons



Glia

Polycomb makes bright loops in neuronal 3D genome





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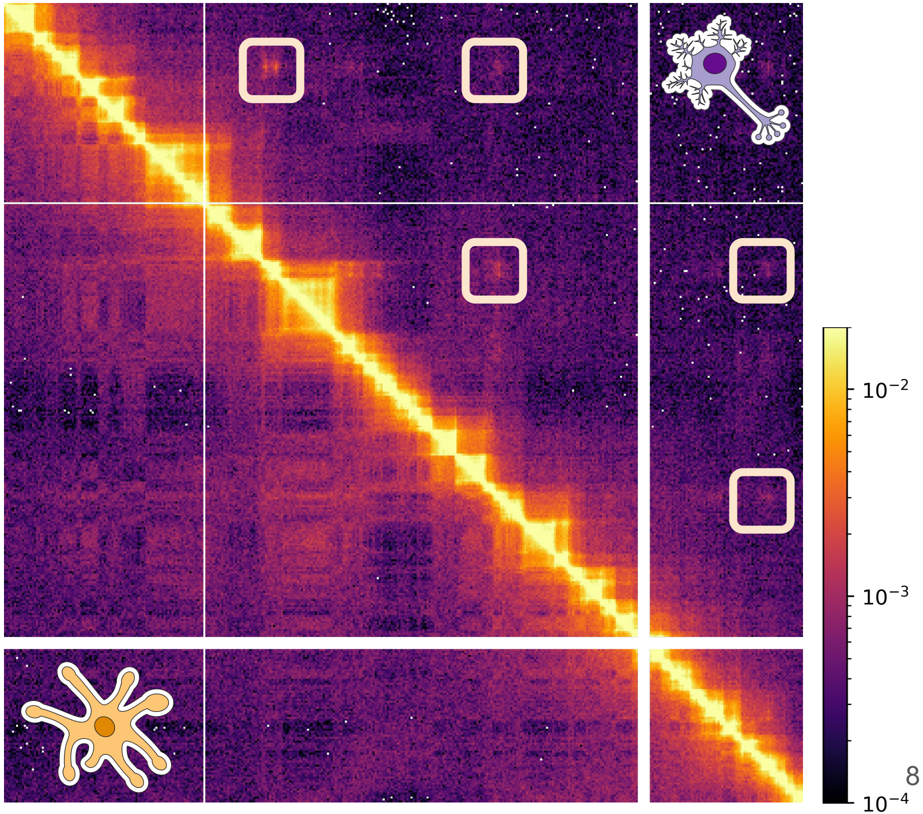
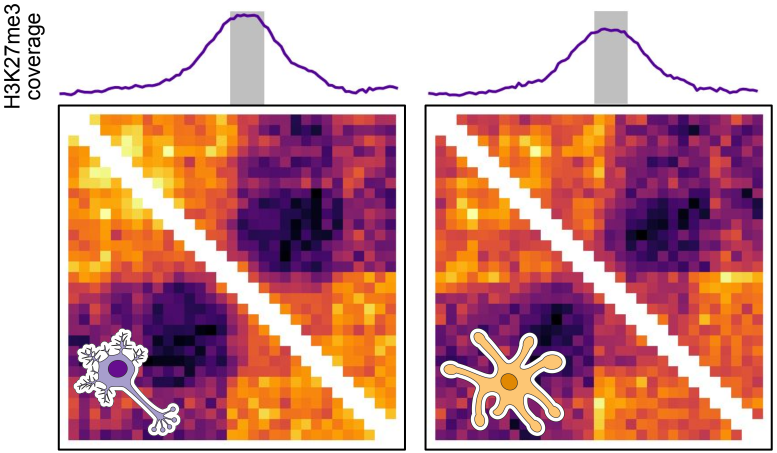


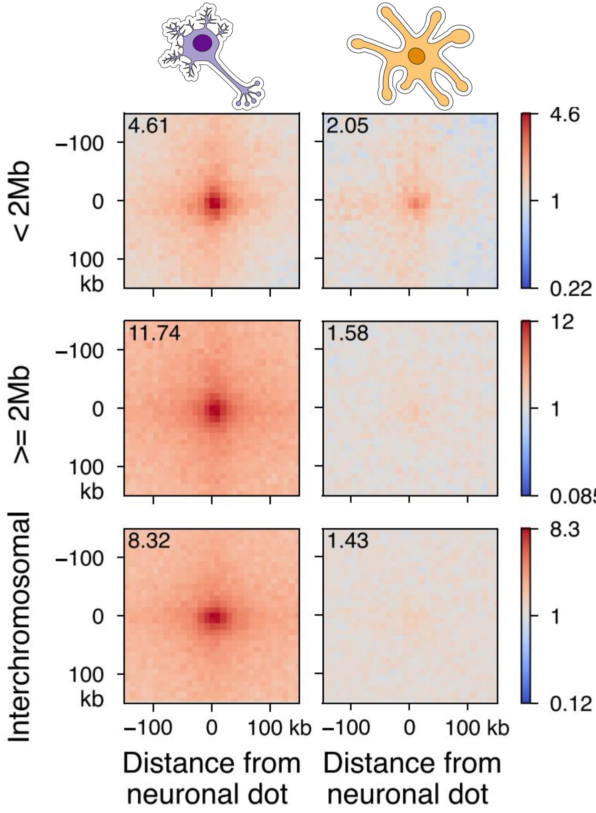
Polycomb loops properties

These loops are:

- **Long-range** (>2Mb)
- Both *cis*- and *trans*-
- Their anchors are enriched with **H3K27me3**

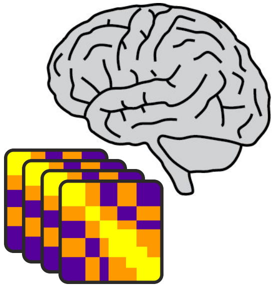


Polycomb loop anchor, ±300 kbp

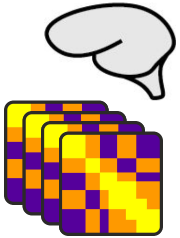


Polycomb in single-cells

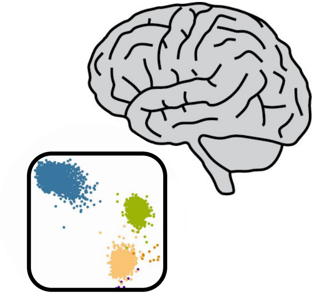
To study Polycomb **across neuronal subtypes**, we performed re-analysis of published human brain “**scHiC**” (sn-m3C-seq) and **scRNAseq** datasets.



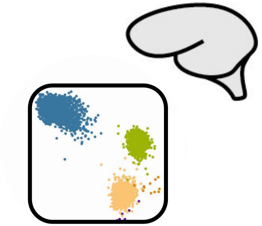
517k cells
W. Tian, ..., J. Ecker, Science 2023



29k cells
M. Heffel, ..., C. Luo, Nature 2024



3.3M cells
K. Siletti, ..., S. Linnarson, Science 2023

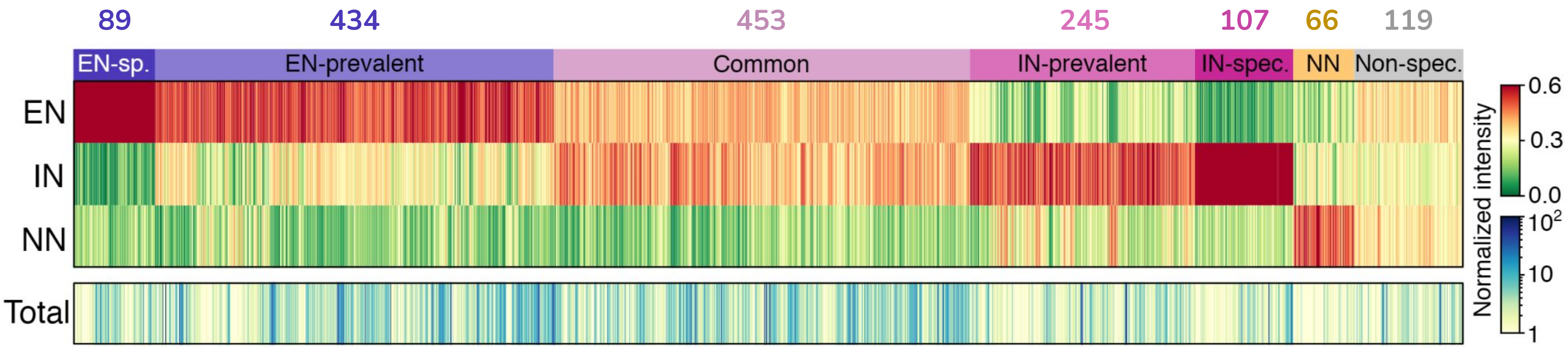


154k cells
C. Herring, ..., R. Lister, Cell 2022

Polycomb loops in neuronal subtypes

We detected **1513** Polycomb loops in total.

Many of them have specificity towards **excitatory** or **inhibitory** neurons.

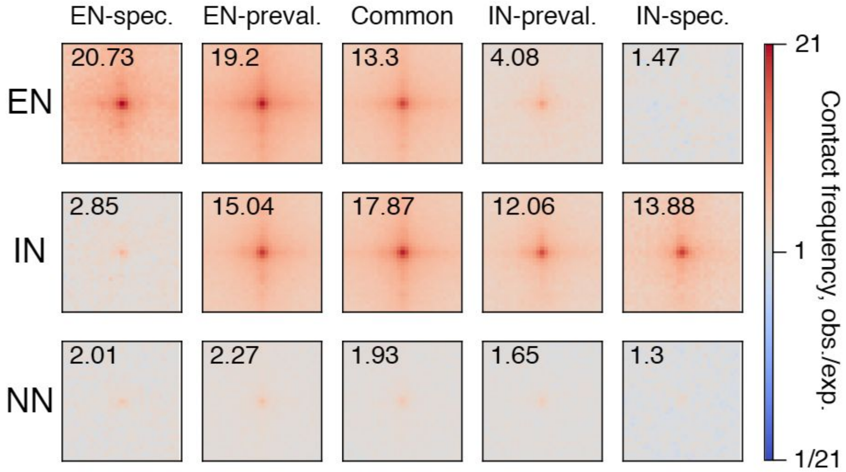


Polycomb loops in neuronal subtypes

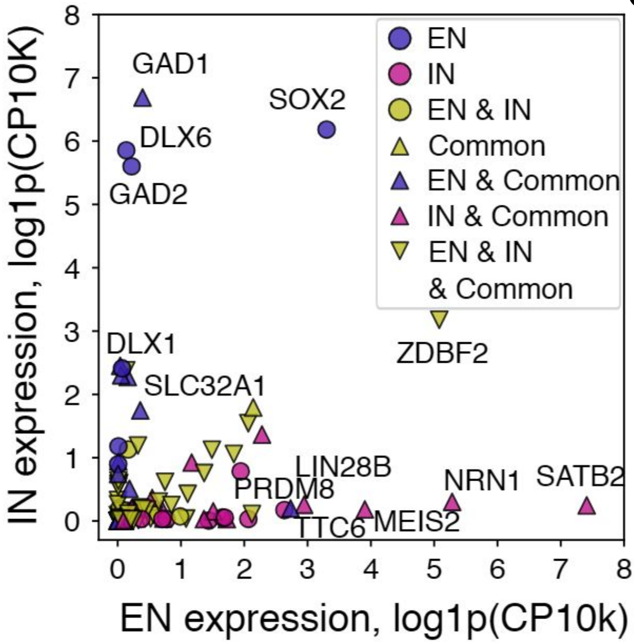
How do they look like?

Do they really affect expression?

Yes



Average snm3C-seq signal from PcG contacts

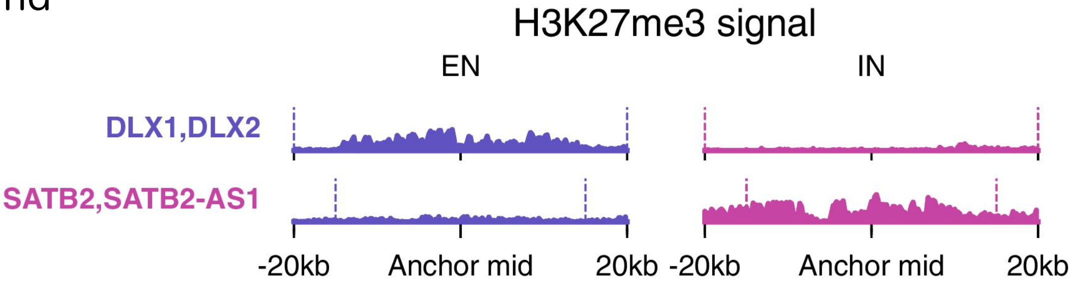


Expression data:
K. Siletti, ..., S. Linnarson, Science 2023

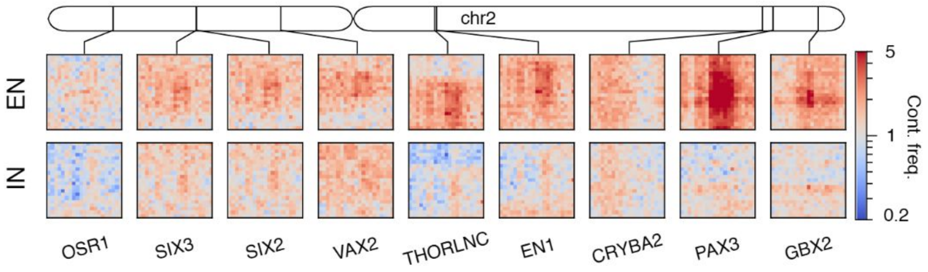
2 examples: DLX1/2 and SATB2

DLX1/2 mostly expressed in EN and SATB2 - in IN.

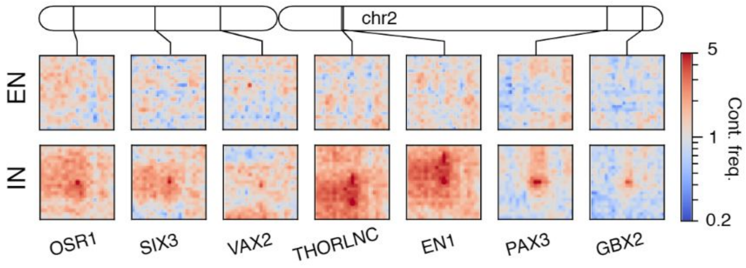
They are all TFs important for **brain development**.



Contacts of ***DLX1/2***

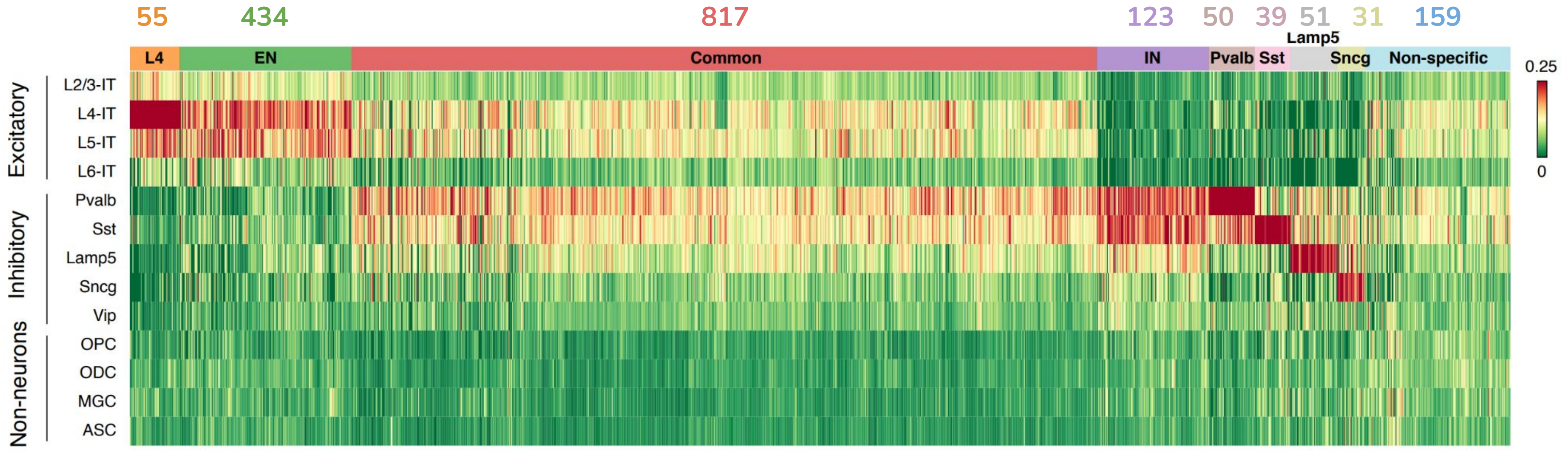


Contacts of ***SATB2***



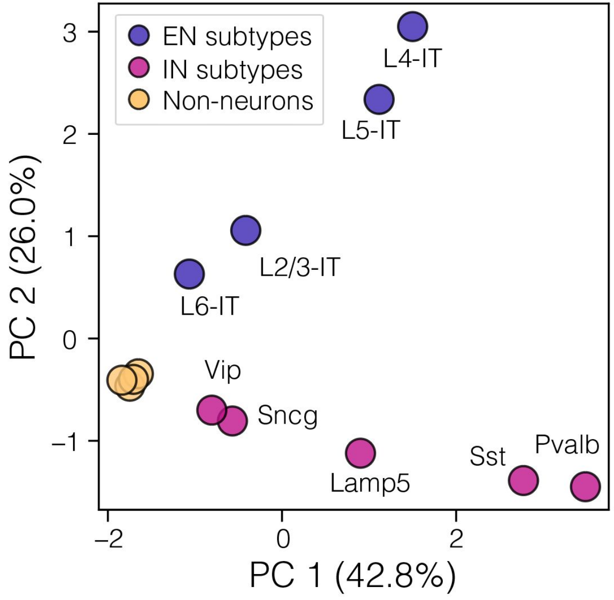
Polycomb loops in neuronal subtypes

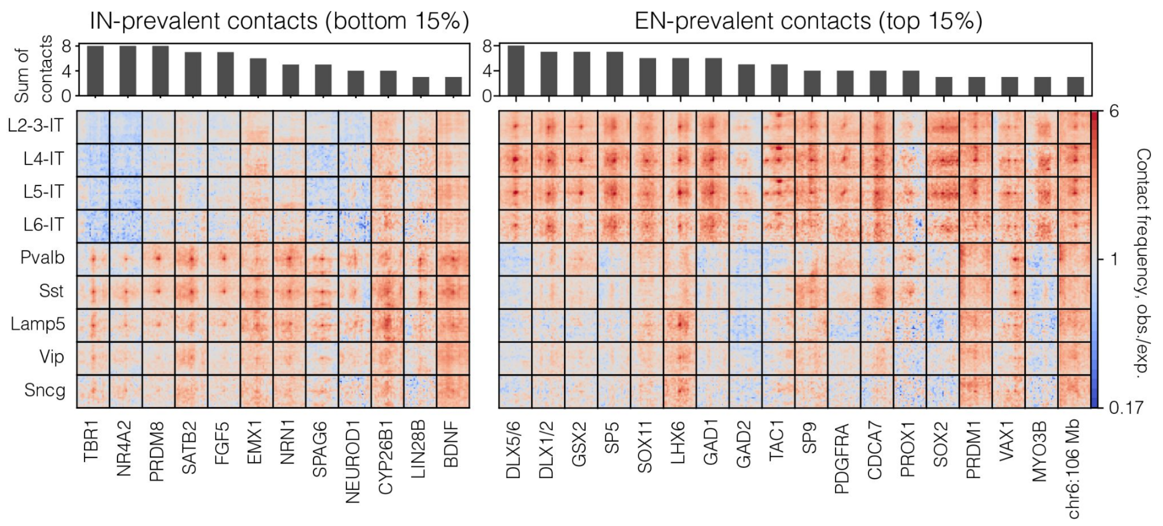
Some of the loops are specific to a certain neuronal subtypes.



Polycomb loops in neuronal subtypes

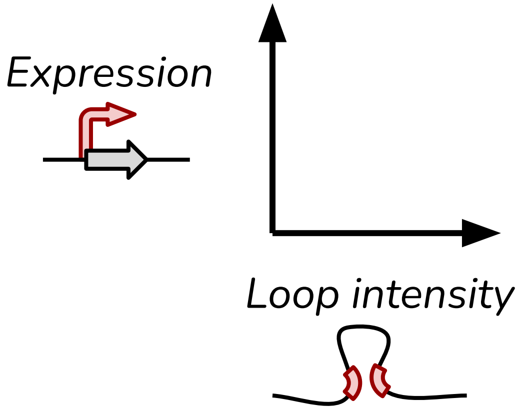
PCA based on the Polycomb loops clearly distinguishes between neuronal subtypes and highlights genes important for excitatory or inhibitory neurons.

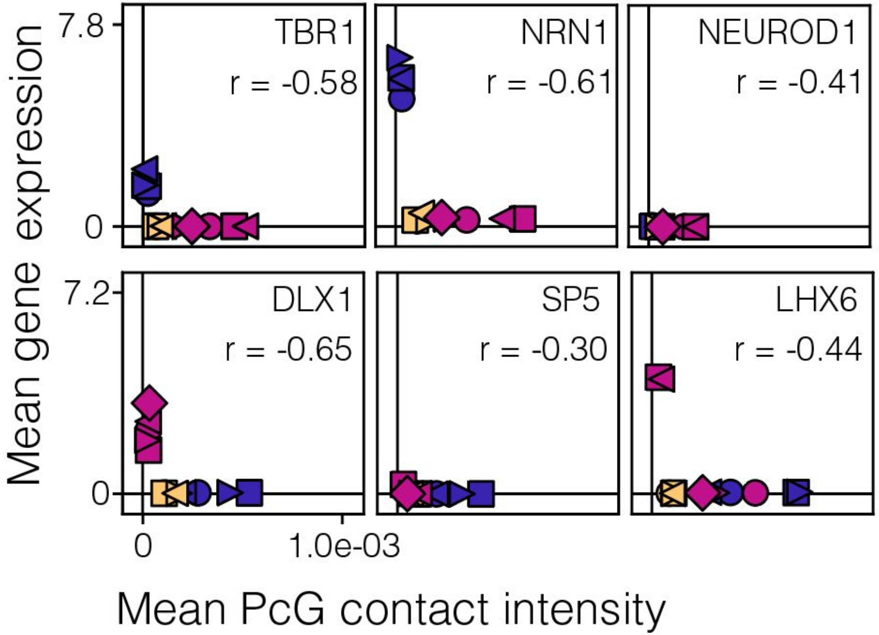




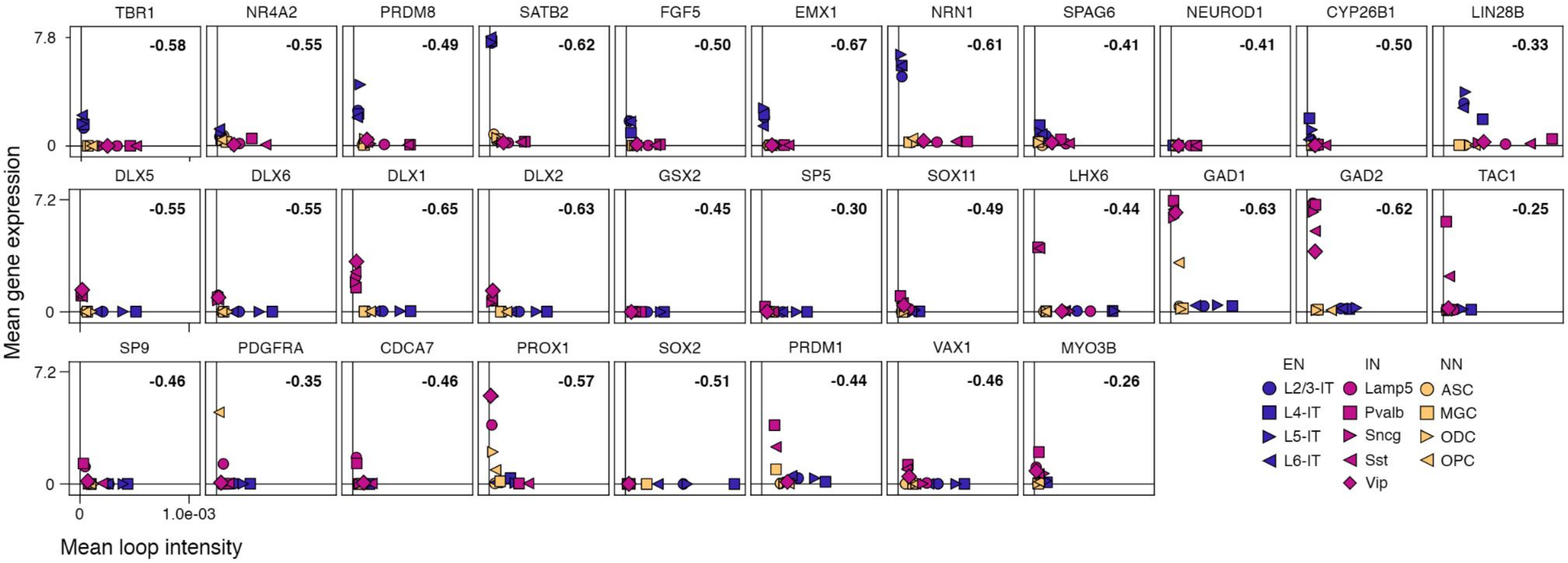
Polycomb loops and gene expression

When loop is present in a certain subtype, the correponding gene is completely **repressed** there

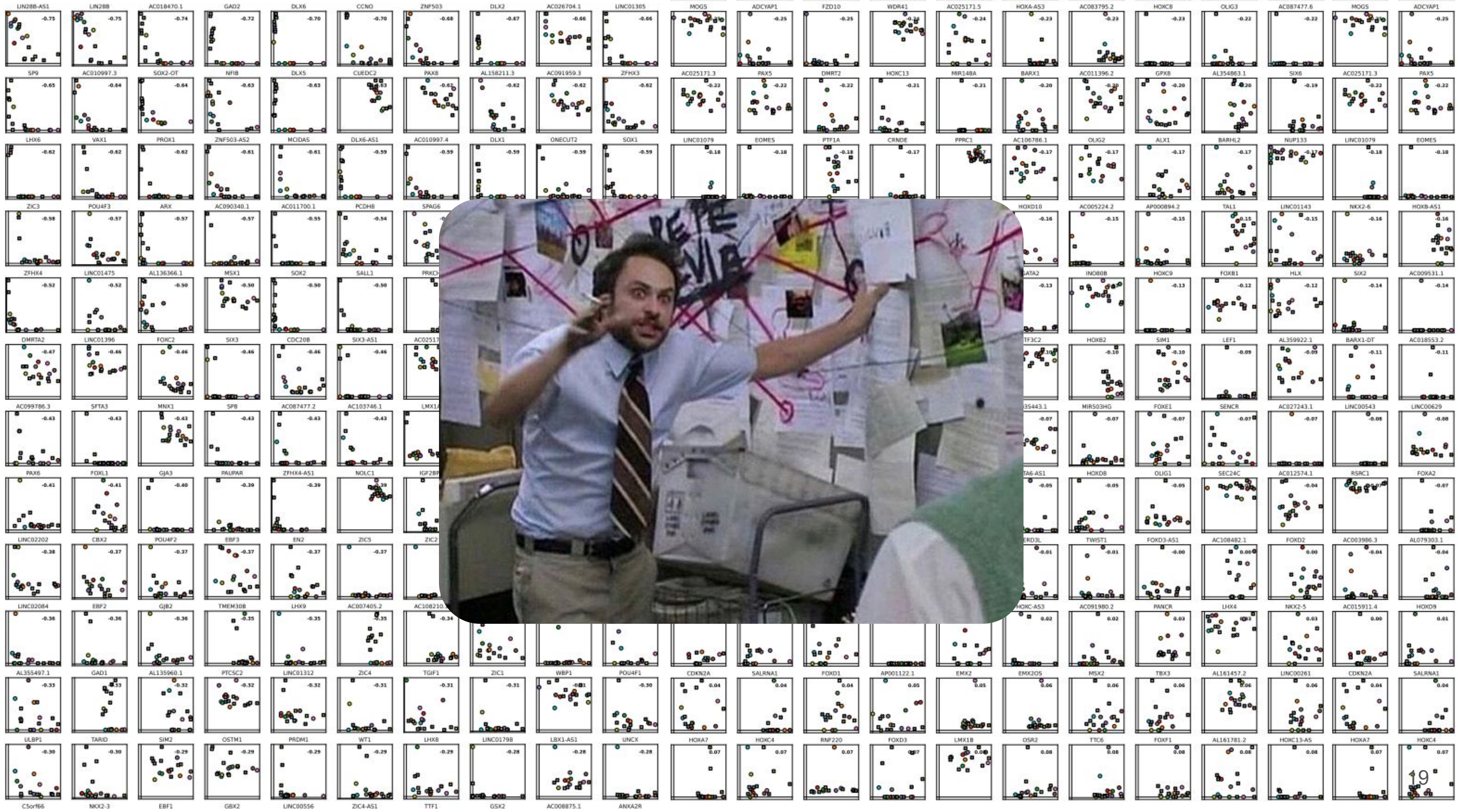




… for many genes

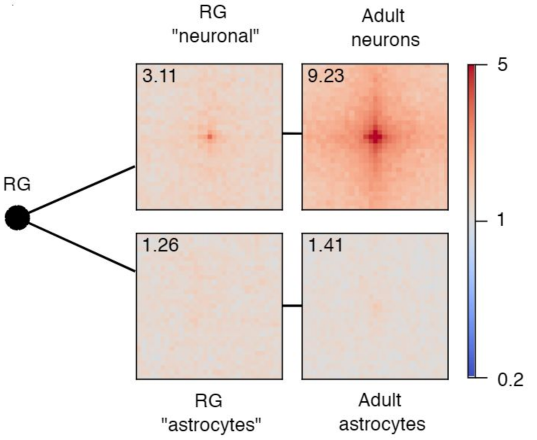


LIN28B-AS1 -0.75
LIN28B -0.75
AC018470.1 -0.74
GAD2 -0.72
DLX6 -0.70
CCNO -0.70
ZNF503 -0.68
DLX2 -0.67
AC026704.1 -0.66
LINC01305 -0.66
LINC02202 -0.38
CBX2 -0.37
POU4F2 -0.37
EBF3 -0.37
SP9 -0.65
AC010997.3 -0.64
SOX2-OT -0.64
NFIB -0.63
DLX5 -0.63
CUEDC2 -0.63
PAX8 -0.62
AL158211.3 -0.62
AC091959.3 -0.62
ZFHX3 -0.62
LINC02084 -0.36
EBF2 -0.36
GJB2 -0.36
TMEM30B -0.35
LHX6 -0.62
VAX1 -0.62
PROX1 -0.62
ZNF503-AS2 -0.61
MCIDAS -0.61
DLX6-AS1 -0.59
AC010997.4 -0.59
DLX1 -0.59
ONECUT2 -0.59
SOX1 -0.59
AL355497.1 -0.33
GAD1 -0.33
AL135960.1 -0.32
PTCSC2 -0.32
ZIC3 -0.58
POU4F3 -0.57
ARX -0.57
AC090340.1 -0.57
AC011700.1 -0.55
PCDH8 -0.54
SPAG6 -0.54
AC009166.1 -0.54
PRDM6 -0.53
CFAP77 -0.52
ULBP1 -0.30
TARID -0.30
SIM2 -0.29
OSTM1 -0.29
ZFHX4 -0.52
LINC01475 -0.52
AL136366.1 -0.50
MSX1 -0.50
SOX2 -0.50
SALL1 -0.50
PRKCH -0.49
NR2E1 -0.49
NBDY -0.48
AC025171.1 -0.48
C5orf66 -0.27
NKX2-3 -0.27
EBF1 -0.27
GBX2 -0.27
DMRTA2 -0.47
LINC01396 -0.46
FOXC2 -0.46
SIX3 -0.46
CDC20B -0.46
SIX3-AS1 -0.46
AC025171.2 -0.46
LINC01833 -0.45
COPZ2 -0.44
AC087477.5 -0.44
MOGS -0.25
ADCYAP1 -0.25
FZD10 -0.25
WDR41 -0.24
AC099786.3 -0.43
SFTA3 -0.43
MNX1 -0.43
SP8 -0.43
AC087477.2 -0.43
AC103746.1 -0.43
LMX1A -0.43
CNPY1 -0.42
LHX5 -0.42
NR2F2-AS1 -0.41
AC025171.3 -0.22
PAX5 -0.22
DMRT2 -0.22
HOXC13 -0.21
PAX6 -0.41
FOXL1 -0.41
GJA3 -0.40
PAUPAR -0.39
ZFHX4-AS1 -0.39
NOLC1 -0.39
IGF2BP3 -0.38
LHX5-AS1 -0.38
AC116667.2 -0.38
NR2F2 -0.38
LINC01079 -0.18
EOMES -0.18
PTF1A -0.18
CRNDE -0.17

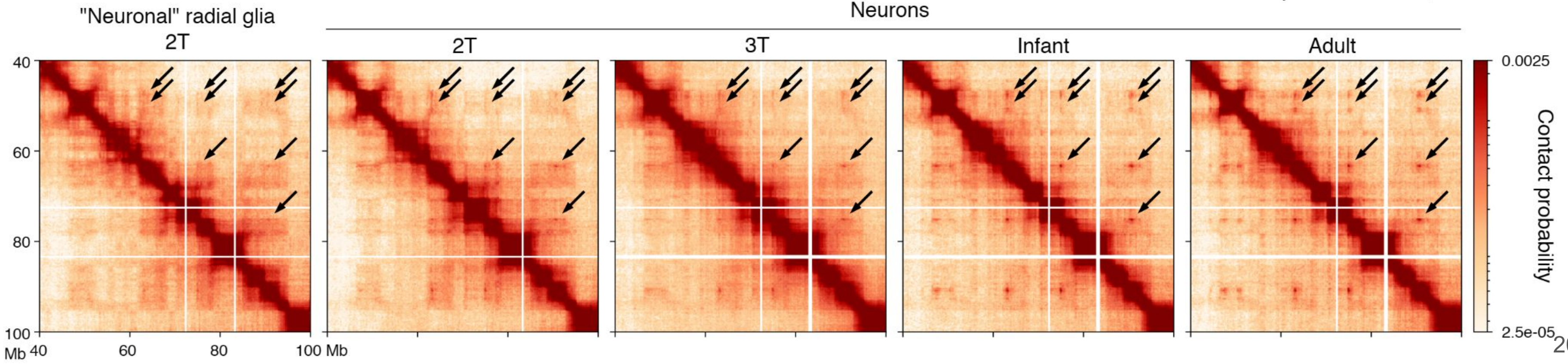


Polycomb loops emerge during development

Polycomb loops are totally **absent** in "astrocytes" radial glia (RG), but present in "neuronal" RG and **gradually appear** during **mid-gestation** (GW18-23)

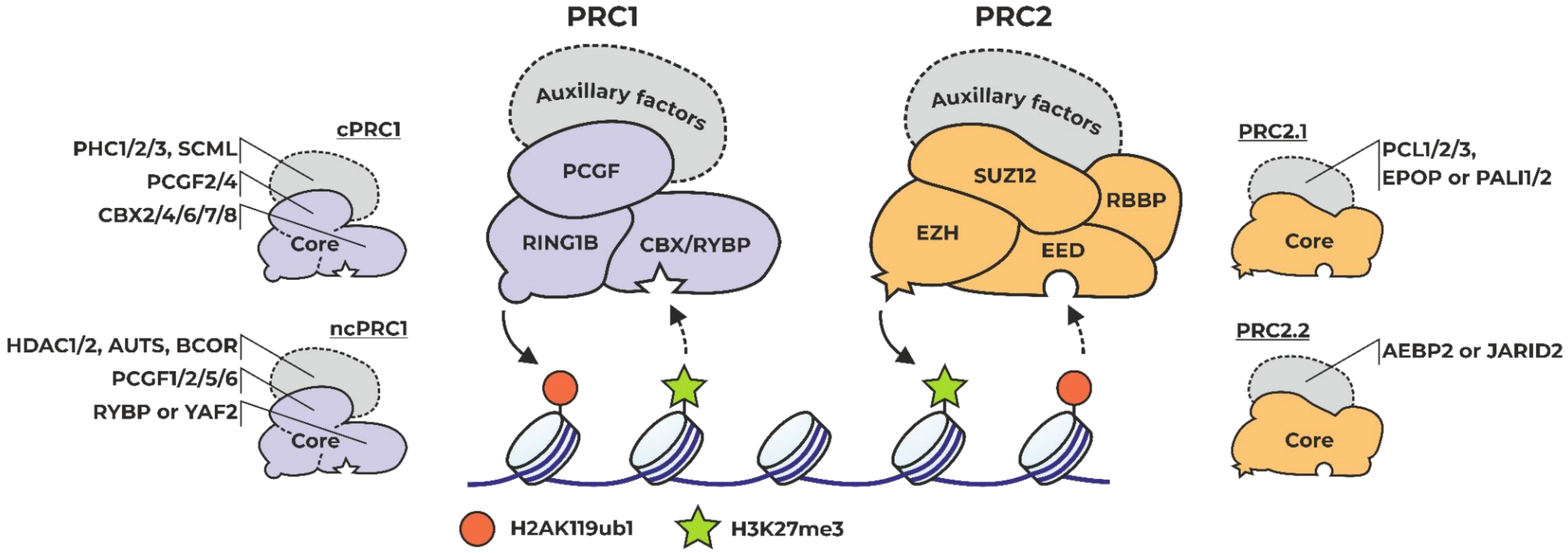


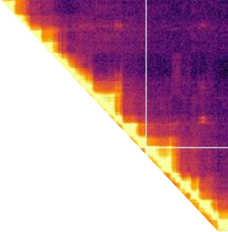
Fetal Hi-C data:
M. Heffel, ..., C. Luo, Nature 2024



Fine-tuning Polycomb activity

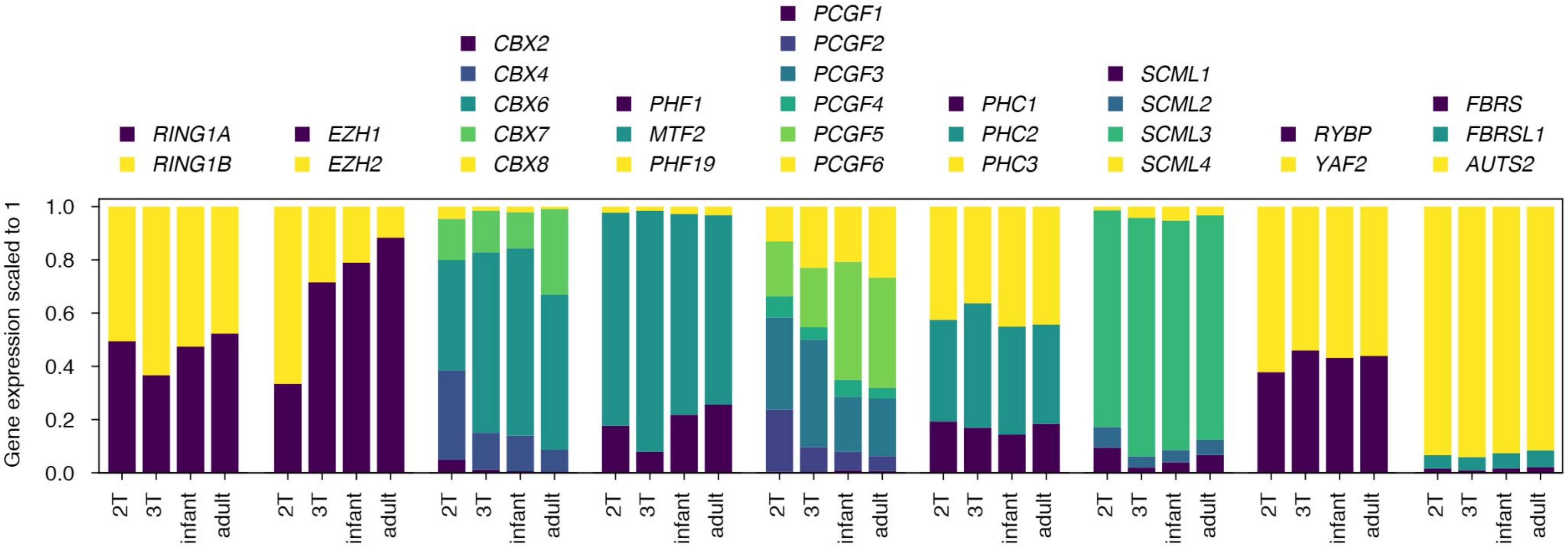
Both PRC1 and PRC2 complexes can be formed from different **paralogs**





Fine-tuning Polycomb activity

Both PRC1 and PRC2 complexes can be formed from different **paralogs**, and their expression alters during development.



Fine-tuning Polycomb activity

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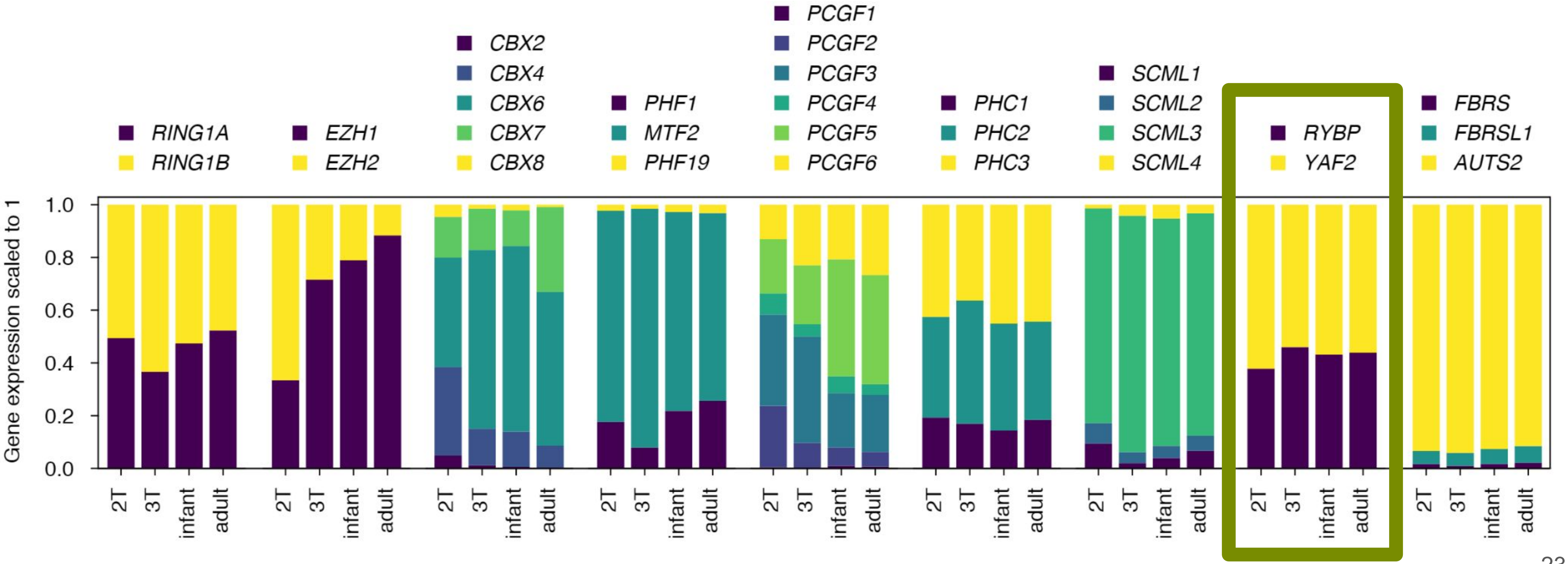
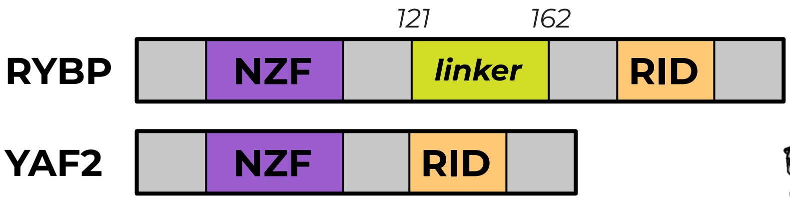
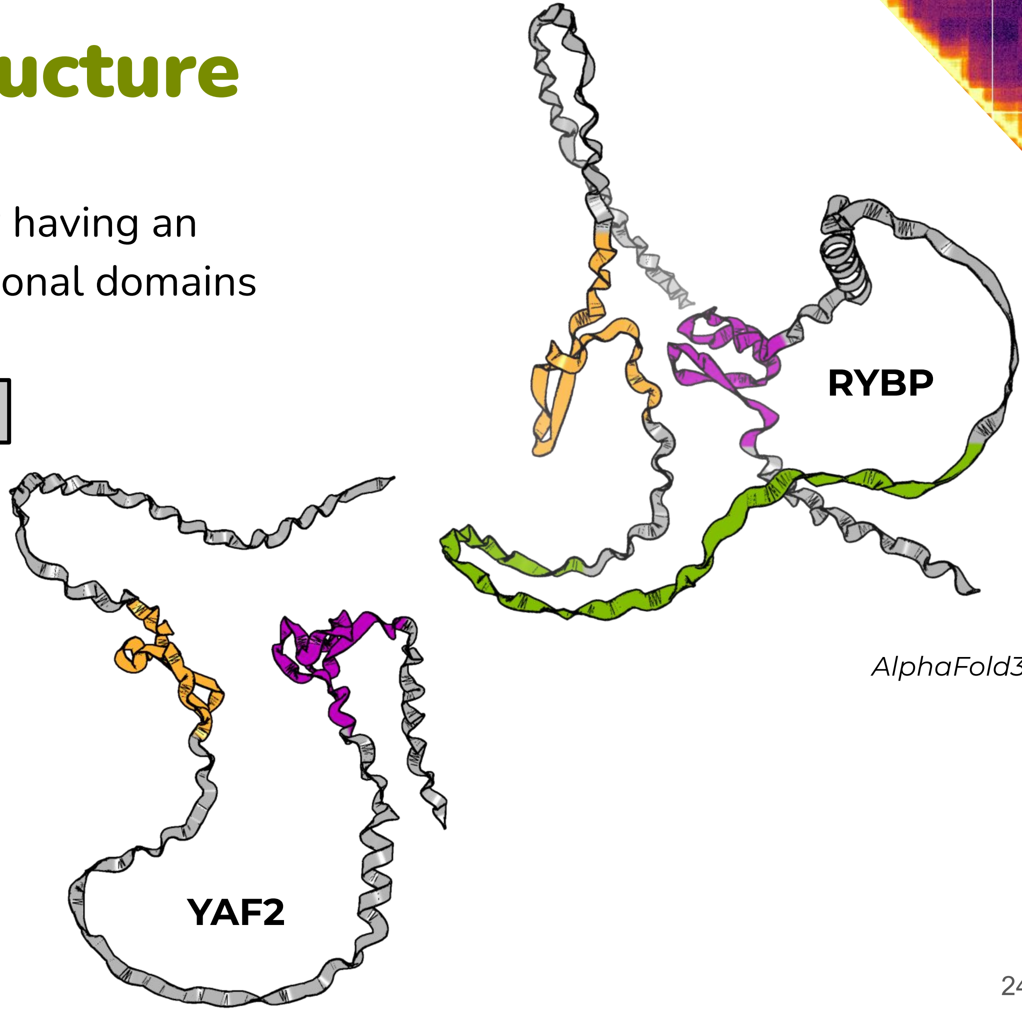


RYBP and YAF2 structure

RYBP and YAF2 are **paralogs** with RYBP having an additional linker part between two functional domains



NZF: Npl4-type zinc finger (H2Aub1 binding)
RID (*cbox*): Ring1B-interacting domain
Linker: Disordered loop domain

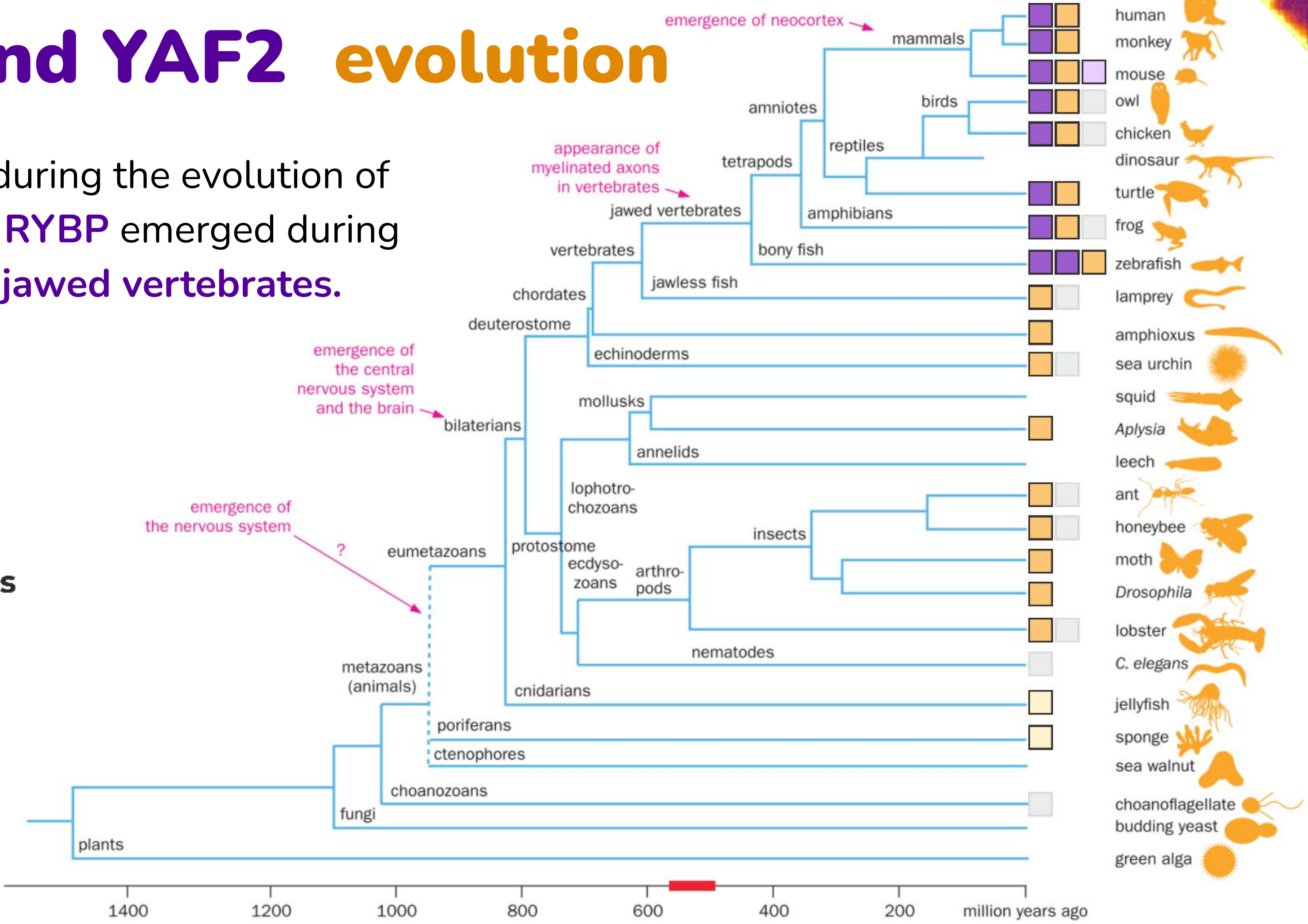


AlphaFold3

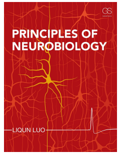
RYBP and YAF2 evolution

YAF2 emerged during the evolution of **bilaterians**, and RYBP emerged during the evolution of **jawed vertebrates.**

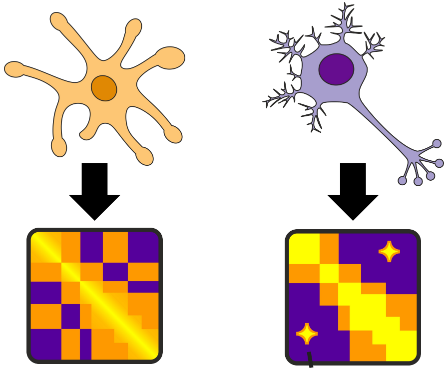
- RYBP
- YAF2
- RYBP pseudo
- YAF2 homologs
- YAF2 domains



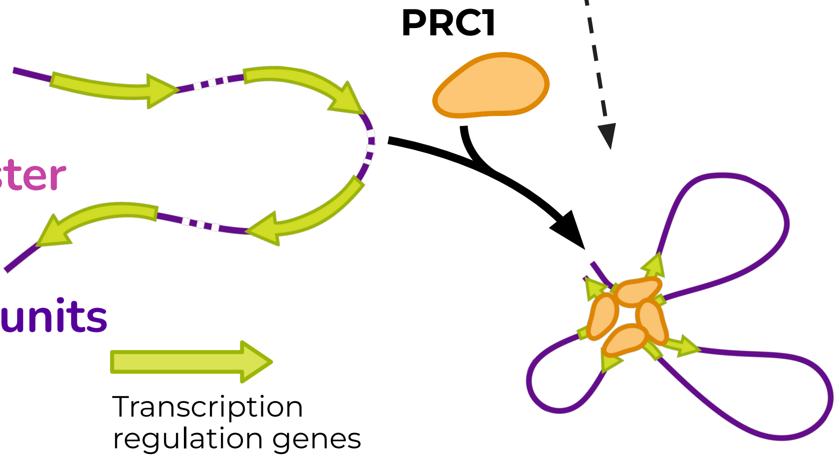
Tree adopted from:
Liqun Luo textbook, 2015



Conclusions



- Polycomb forms **long-range** loops in **neurons**
- Many Polycomb loops are **specific** to a certain neuronal types and **subtypes**
- Polycomb **restricts expression** of certain genes in a cell-type-dependent manner. For instance, it represses *DLX1/2* in EN and SATB2 in IN
- Polycomb regulatory landscape is established in the brain during **2nd trimester**
- It's activity can be tweaked by **differential usage of** interchangeable **subunits**



Acknowledgments

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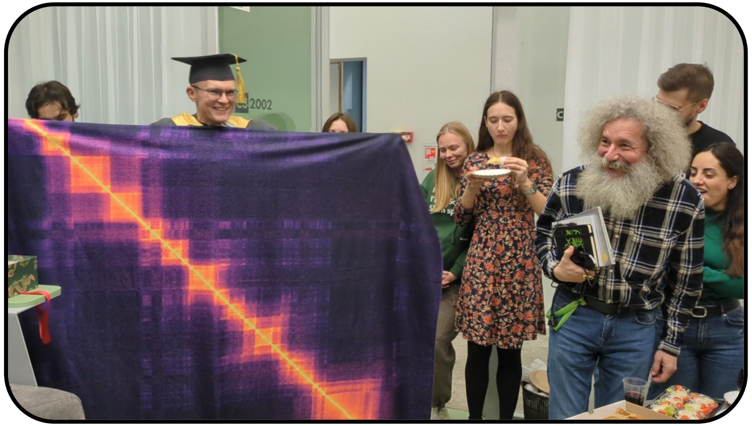
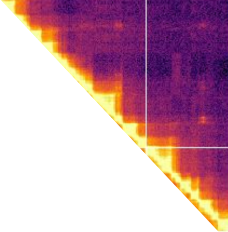
- Maria Molodova
- Anastasia Soldatenkova
- Ekaterina Kuznechenkova

... and many others!

Russian Science Foundation
IDB RAS Core Centrum
Skoltech Genomics Core Facility
Skoltech BioImaging and Spectroscopy Core Facility
IGB RAS CPGE GTB
National BioService Russian Biospecimen CRO
NIMH Repository & Genomics Resource





If you are interested...

Check out our latest preprint!

Ultra-long-range Polycomb-coupled interactions underlie subtype identity of human cortical neurons

Ilya A. Pletenev, Nikita Vaulin, Maria N. Molodova, Ekaterina Kuznechenkova, Anastasia Soldatenkova, Olga I. Efimova, Anna V. Tvorogova, Philipp Khaitovich, Sergey V. Razin, Sergey V. Ulianov, Ekaterina E. Khrameeva

doi: https://doi.org/10.1101/2025.11.05.686502

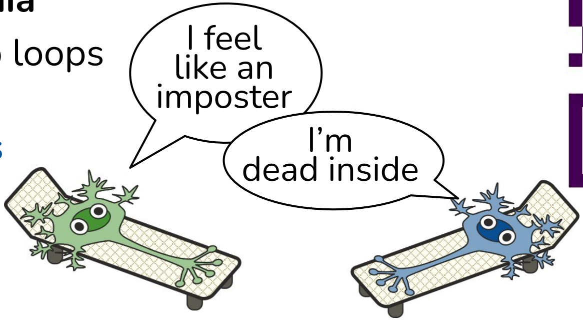
Or our previous paper on Polycomb in the brain

Extensive long-range polycomb interactions and weak compartmentalization are hallmarks of human neuronal 3D genome

Ilya A. Pletenev[1,†], Maria Bazarevich[1,†], Diana R. Zagirova[1,2,†], Anna D. Kononkova[1], Alexander V. Cherkasov[1], Olga I. Efimova[3], Eugenia A. Tiukacheva[4,5,6,7,8], Kirill V. Morozov[1], Kirill A. Ulianov[1], Dmitriy Komkov[9], Anna V. Tvorogova[9], Vera E. Golimbet[10], Nikolay V. Kondratyev[10], Sergey V. Razin[5,8], Philipp Khaitovich[3], Sergey V. Ulianov[5,8,*] and Ekaterina E. Khrameeva[1,*]

And also we are studying:

- Chromatin structure in **schizophrenia**
- **Automatic** annotation of Polycomb loops
- Differences between **iPSC-derived** and **post-mortem neurons**
- **Aging**



all the links are here

nvaulin.github.io/BrainEpiGenome2026